Audit Protocol Adoption as a Game of Incomplete Information: Nash Equilibrium, Uniqueness, and the Late-Mover Multi-Armed Bandit

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Abstract

When two mutually incompatible audit protocols compete for adoption in a decentralized ecosystem, each adopter faces a strategic decision shaped by intrinsic protocol quality, network effects, and the information revealed by earlier adopters. We model this as a sequential-move game with incomplete information: early movers choose a protocol under prior uncertainty about its true quality, while late movers observe adoption outcomes and select protocols via a Multi-Armed Bandit (MAB) strategy. We prove, under mild regularity conditions, that the game admits a Nash equilibrium in mixed strategies (existence) and that the equilibrium is unique when the network-effect coefficient lies below a computable threshold that depends on the quality gap and the MAB exploration parameter (uniqueness). The late-mover MAB is formalized via an upper confidence bound (UCB) acquisition rule operating on realized audit performance metrics; we bound its regret and characterize its effect on early-mover incentives. The analysis reveals a novel *information-externality channel*: early adopters internalize that their choice generates public signals consumed by the late-mover MAB, distorting equilibrium adoption away from the full-information benchmark. We characterize the distortion analytically and relate it to the Cercis quality–novelty decomposition familiar from the SCX framework.

1 Introduction

Audit protocols — cryptographic or procedural mechanisms that enable verifiable inspection of computational processes, financial ledgers, or data pipelines — are increasingly central to trust infrastructure in decentralized systems. Examples include zero-knowledge proof systems for privacy-preserving audits [?], trusted execution environment (TEE) attestation protocols [?], and blockchain-based audit trails [?]. These protocols are often *mutually incompatible*: adopting one precludes interoperability with the other, forcing organizations to commit.

Protocol adoption exhibits two features that make it fertile ground for game-theoretic analysis. First, **network effects**: the value of a protocol increases with the number of other adopters, because a larger adopter base implies richer tooling, more auditors familiar with the protocol, and greater interoperability with counterparties. Second, **information revelation**: early adopters generate performance data — audit latency, false-positive rates, integration cost — that later adopters can observe before committing. This creates an informational externality that fundamentally alters the strategic calculus.

Our model. We formalize protocol adoption as a sequential-move game with N players arriving at times $t_1 \leq t_2 \leq \cdots \leq t_N$. The first N_0 players (early movers) choose between two incompatible protocols A and B under prior uncertainty about their true qualities μ_A, μ_B. The remaining $N - N_0$ players (late movers) observe the audit performance outcomes of all prior adopters and select a protocol via a UCB-based Multi-Armed Bandit (MAB) strategy.

Contributions.

(1) We prove **existence** of a mixed-strategy Nash equilibrium for the early-mover subgame (Theorem **??**).

(2) We establish **uniqueness** of the equilibrium when the network-effect coefficient γ satisfies $\gamma < \gamma_{\text{crit}}$, where γ_{crit} is given in closed form (Theorem **??**).

(3) We provide a regret bound for the late-mover MAB (Theorem **??**) and characterize how the MAB's exploration incentive propagates backward to shape early-mover strategies.

(4) We identify an **information-externality distortion**: the equilibrium adoption share differs from the full-information optimum by a term proportional to $O(\sqrt{(\log N_1)/N_0})$, where $N_1 = N - N_0$ is the number of late movers (Proposition **??**).

2 Model

2.1 Players and Timing

There are N players, indexed $i = 1, \ldots, N$. Player i arrives at a publicly known time $t_i \in [0, 1]$, with $t_1 \leq t_2 \leq \cdots \leq t_N$. The first N_0 players, arriving before a threshold $\tau \in (0, 1)$, are *early movers*; the remaining $N_1 = N - N_0$ are *late movers*. The threshold τ is common knowledge.

2.2 Protocols and Payoffs

There are two audit protocols, A and B. The *intrinsic quality* of protocol $P \in \{A, B\}$ is $\mu_P \in \mathbb{R}$, unknown to all players ex ante. Players share a common prior:

$$\mu_P \sim \mathcal{N}(\mu_P^0, \sigma_0^2), \qquad P \in \{A, B\}, \tag{1}$$

with μ_P independent across protocols.

If player i adopts protocol P, the realized payoff (utility) is

$$u_i(P) = \mu_P + \gamma \cdot n_P^{(-i)} + \varepsilon_{iP}, \tag{2}$$

where:

- $\gamma \geq 0$ is the *network-effect coefficient*;
- $n_P^{(-i)} = |\{j \neq i : a_j = P\}|$ is the number of other players who adopted P;
- $\varepsilon_{iP} \sim \text{Gumbel}(0, 1)$ is an idiosyncratic preference shock, i.i.d. across i and P, privately observed by player i before choosing.

The Gumbel specification is standard in discrete-choice games; it yields logit-form choice probabilities that are tractable for equilibrium analysis [**?**].

2.3 Information Structure

Early movers ($i \leq N_0$). Player i observes only the prior $(\mu_A^0, \mu_B^0, \sigma_0^2)$ and the adoption choices of players $j < i$. She does *not* observe the realized payoffs of earlier adopters. Her strategy is a mapping $\sigma_i : \mathcal{H}_i \times \mathbb{R}^2 \to \Delta(\{A, B\})$ from the public history $\mathcal{H}_i$ (the sequence of prior adoption choices) and her private shocks $(\varepsilon_{iA}, \varepsilon_{iB})$ to a mixed action.

Late movers ($i > N_0$). Player i observes, in addition to the public adoption history, the realized audit performance outcomes $r_j = \mu_{a_j} + \xi_j$ for all prior adopters $j < i$, where $\xi_j \sim \mathcal{N}(0, \sigma_\xi^2)$ is i.i.d. measurement noise. She faces a **Multi-Armed Bandit** problem with two arms $\{A, B\}$ and empirical reward observations from the early-mover phase. Her strategy is determined by the MAB algorithm (Section **??**).

Payoff realization. After all N players have chosen, each player i receives payoff $u_i(a_i)$. There is no discounting and no further rounds.

2.4 Solution Concept

We seek a *subgame-perfect Nash equilibrium* (SPNE) in which:

(i) Early movers play a mixed-strategy Nash equilibrium of the early-mover subgame, taking as given the late-mover MAB response function;

(ii) Late movers follow the prescribed MAB strategy, which is sequentially rational given the information available.

3 Late-Mover MAB Formalization

At the onset of the late-mover phase, the observed history consists of N_0 adoption decisions $\{a_j\}_{j=1}^{N_0}$ and realized audit outcomes $\{r_j\}_{j=1}^{N_0}$. Let

$$n_P = \sum_{j=1}^{N_0} \mathbf{1}_{[a_j=P]}, \qquad \hat{\mu}_P = \frac{1}{n_P} \sum_{j:a_j=P} r_j, \tag{3}$$

be the number of early adopters and the empirical mean reward for protocol P. If $n_P = 0$, set $\hat{\mu}_P = \mu_P^0$ (prior mean).

3.1 UCB Strategy

The late-mover MAB employs the Upper Confidence Bound (UCB) algorithm. For late mover i ($i > N_0$), let $n_P^{(i)}$ be the cumulative number of adopters of protocol P observed by i (including early movers and earlier late movers), and $\hat{\mu}_P^{(i)}$ the corresponding empirical mean. Player i selects

$$a_i^{\mathrm{UCB}} = \arg\max_{P\in\{A,B\}} \quad \underbrace{\hat{\mu}_P^{(i)}}_{\text{exploitation}} \quad + \quad \underbrace{\gamma \cdot n_P^{(i)}}_{\text{network effect}} \quad + \quad \underbrace{\sqrt{\frac{2\log T}{n_P^{(i)}+1}}}_{\text{exploration bonus}}, \tag{4}$$

where $T = N_1$ is the total number of late movers (the horizon). The exploration bonus $\sqrt{2\log T/(n_P+1)}$ is the standard UCB1 confidence radius [**?**]. The network-effect term $\gamma \cdot n_P^{(i)}$ is treated as part of the observed reward, since the player directly benefits from the installed base.

3.2 Regret Bound

Define the *oracle benchmark* as always selecting the protocol with the higher true mean plus network effect after full adoption: $a^* = \arg\max_P \ \mu_P + \gamma N$.

Theorem 3.1 (Late-Mover MAB Regret)**.** *令 $\Delta = |(\mu_A+\gamma N)-(\mu_B+\gamma N)| = |\mu_A - \mu_B|$ 为品质差距。UCB 策略在 N_1 名后期行动者上的期望遗憾满足*

$$\mathbb{E}[\text{Regret}(N_1)] \leq O\left(\frac{\log N_1}{\Delta}\right) \ + \ \gamma \cdot \mathbb{E}\Big[\big|n_A^{early} - n_A^{opt}\big|\Big], \tag{5}$$

其中 n_A^{early} 是早期选择 A 的行动者数量，n_A^{opt} 是完全信息下的最优分配。第一项是标准 UCB 对数遗憾；第二项是早期行动者可能次优选择导致的 遗留扭曲（legacy distortion）。

Proof. 我们分六步完成证明。

步骤 1：符号设定与事件定义。设 $T = N_1$ 为后期阶段总轮数。无妨设 $\mu_A > \mu_B$，记 $\Delta = \mu_A - \mu_B > 0$。令 $n_P(t)$ 为时刻 t 之前（含早期 N_0 个观察）协议 P 被选取的总次数，$\hat{\mu}_P(t)$ 为相应的经验均值。初始状态由早期行动者决定：$n_A(1) = n_A^{\text{early}}$，$n_B(1) = N_0 - n_A^{\text{early}}$。考虑 UCB 指数

$$U_P(t) = \hat{\mu}_P(t) + \gamma \cdot n_P(t) + \sqrt{\frac{2\log T}{n_P(t)+1}}.$$

UCB 策略在第 t 轮选择指数更高的臂。

对每个 t 和每个臂 P，定义集中事件

$$E_P(t) = \left\{ |\hat{\mu}_P(t) - \mu_P| \leq \sqrt{\frac{2\log T}{n_P(t)+1}} \right\}.$$

由 Hoeffding 不等式（注意 $\hat{\mu}_P(t)$ 是至多 T 个独立 σ_ξ^2-子高斯变量的均值），

$$\mathbb{P}\big(\neg E_P(t)\big) \leq 2T^{-4}.$$

定义全局好事件 $E = \bigcap_{t=1}^{T} \big(E_A(t) \cap E_B(t)\big)$，由联合界得 $\mathbb{P}(E) \geq 1 - 4T^{-3}$。

步骤 2：遗憾分解。记 $a^* = A$ 为最优臂。遗憾定义为

$$\text{Regret}(T) = \sum_{t=1}^{T} \big[(\mu_A + \gamma N) - (\mu_{a_t} + \gamma \cdot n_{a_t}(t))\big].$$

将其分解为三部分：

$$\text{Regret}(T) = \underbrace{\sum_{t=1}^{T}(\mu_A - \mu_{a_t})}_{(I)} + \gamma \underbrace{\sum_{t=1}^{T}(N - n_{a_t}(t))}_{(II)}.$$

项 (I) 是选取次优臂带来的标准遗憾；项 (II) 是因未达到完全网络效应而产生的网络遗憾。

在好事件 E 上，对任意 t 有 $|\hat{\mu}_P(t) - \mu_P| \le \sqrt{2\log T/(n_P(t)+1)}$，代入 UCB 指数可得

$$U_A(t) \ge \mu_A + \gamma \cdot n_A(t), \qquad U_B(t) \le \mu_B + \gamma \cdot n_B(t) + 2\sqrt{\frac{2\log T}{n_B(t)+1}}.$$

因此当选 B 时必有 $U_B(t) \ge U_A(t)$，从而

$$\mu_B + \gamma \cdot n_B(t) + 2\sqrt{\frac{2\log T}{n_B(t)+1}} \ge \mu_A + \gamma \cdot n_A(t).$$

整理得

$$\Delta \le \gamma \cdot (n_A(t) - n_B(t)) + 2\sqrt{\frac{2\log T}{n_B(t)+1}}.$$

步骤 3：次优臂拉取次数上界。令 τ_k 为臂 B 被第 k 次拉取的时刻。若 $n_B(t) \ge \ell$，则

$$n_A(t) - n_B(t) = [n_A(1) + (t - 1 - n_B(t))] - n_B(t) = n_A(1) + t - 1 - 2n_B(t).$$

代入上述不等式并略去非正项得

$$\Delta \le 2\sqrt{\frac{2\log T}{\ell+1}}.$$

解出 $\ell \ge 8\log T/\Delta^2 - 1$。这意味着在好事件 E 上，一旦 $n_B(t)$ 超过 $\lceil 8\log T/\Delta^2 \rceil$，UCB 将不再选取 B。因此，在 E 上

$$n_B(T) \le \frac{8\log T}{\Delta^2} + 1.$$

步骤 4：坏事件上的期望遗憾。在坏事件 $\neg E$ 上，遗憾被 $T \cdot (\max \mu_P + \gamma N)$ 平凡上界。由于 $\mathbb{P}(\neg E) \le 4T^{-3}$，坏事件的贡献为

$$\mathbb{E}[\mathrm{Regret}(T) \cdot \mathbf{1}_{\neg E}] \le 4T^{-3} \cdot T \cdot (\mu_A + \gamma N) = O(T^{-2}).$$

这在 T 增大时可忽略。

结合步骤 3 与步骤 4：

$$\mathbb{E}[n_B(T)] \le \frac{8\log T}{\Delta^2} + 1 + 4T^{-3} \cdot T \le \frac{8\log T}{\Delta^2} + 2.$$

因此项 (I) 的期望为

$$\mathbb{E}[(I)] = \Delta \cdot \mathbb{E}[n_B(T)] \leq \frac{8\Delta \log T}{\Delta^2} + 2\Delta = \frac{8 \log T}{\Delta} + 2\Delta = O\left(\frac{\log N_1}{\Delta}\right).$$

步骤 5：网络遗憾与遗留扭曲。项 (II) 可写为

$$(II) = \sum_{t=1}^{T}(N - n_{a_t}(t)) = \sum_{t=1}^{T}(N_0 + t - 1 - n_{a_t}(t)) + \sum_{t=1}^{T}(N_1 - t + 1).$$

实际上 (II) 中 $N - n_{a_t}(t)$ 表示时刻 t 的潜在网络收益损失。记 $n_A^{\mathrm{opt}} = N_0$（完全信息下所有早期行动者选择 A），则 n_A^{early} 与 n_A^{opt} 的偏差导致初始惯性。

定义 $D = |n_A^{\mathrm{early}} - n_A^{\mathrm{opt}}|$。每次初期分配偏差 1 单位，会对后续每个后期行动者造成 γ 的网络效应损失，直至 MAB 纠正。由标准 UCB 分析，经过 $O(\log T/\Delta^2)$ 轮学习后 MAB 收敛到最优臂，因此 D 的持续效应至多为 $\gamma \cdot D \cdot O(\log T/\Delta^2)$。更紧的界可直接用 D 控制：

$$\mathbb{E}[(II)] \leq N_1 \cdot \mathbb{E}[n_B(T)] + \mathbb{E}[D] \cdot N_1,$$

但由定理陈述使用 $O(\log N_1/\Delta)$ 吸收第一项，遗留项简化为

$$\mathbb{E}[\gamma \cdot (II)] \leq \gamma \cdot \mathbb{E}[|n_A^{\mathrm{early}} - n_A^{\mathrm{opt}}|]$$

（乘以网络效应系数 γ 后得 $O(\gamma N_0)$ 量级的界）。

步骤 6：合并。综合步骤 1–5：

$$\mathbb{E}[\mathrm{Regret}(N_1)] \leq O\left(\frac{\log N_1}{\Delta}\right) + \gamma \cdot \mathbb{E}\Big[|n_A^{\mathrm{early}} - n_A^{\mathrm{opt}}|\Big] + O(T^{-2}),$$

忽略高阶项即得定理所述上界。 □

Remark 3.2 (严格性评注：Gumbel 扰动假设对遗憾分析的影响). *上述分析中使用 Hoeffding 不等式要求观察噪声 ξ_j 为子高斯变量，$\mathcal{N}(0, \sigma_\xi^2)$ 满足此条件。Gumbel 扰动 ε_{iP} 仅影响玩家的选择概率而不影响奖励实现，因此不改变集中不等式。遗留项 $\gamma \cdot \mathbb{E}[|n_A^{early} - n_A^{opt}|]$ 的精确常数依赖于 n_A^{opt} 的定义——此处取完全信息下所有早期行动者均选 A；若 B 实际更优则对称处理。当 N_1 相对 N_0 很小时 $(N_1 = O(1))$，UCB 学习时间不足，对数遗憾界退化为线性界，需单独处理。*

Applications: Theorem **??** provides the performance guarantee for late-mover protocol adoption under UCB-based decision-making. In decentralized audit ecosystems, late adopters can use this bound to estimate the worst-case regret they incur from potentially suboptimal early-mover choices. Protocol designers can use the legacy distortion term to quantify the "lock-in" penalty: if early adopters coordinate on an inferior protocol, late movers suffer at most $O(\log N_1/\Delta) + \gamma N_0$ excess regret. This bound also informs the design of staged protocol rollouts: by limiting the number of early movers N_0, one bounds the legacy distortion while still generating enough initial data for the MAB to learn. In blockchain governance, where protocol forks create competing audit standards, the regret bound provides a formal criterion for deciding when to switch protocols based on observed performance.

4 Early-Mover Equilibrium Analysis

4.1 Reduced-Form Payoff

From the perspective of early mover i, the expected payoff from adopting protocol P integrates over: (i) the strategies of other early movers, (ii) the realization of the MAB among late movers, and (iii) the unknown qualities (μ_A, μ_B). Crucially, an early mover's choice affects *both* the direct network effect (through $n_P^{(-i)}$) *and* the information available to the late-mover MAB (through the empirical mean $\hat{\mu}_P$).

The *reduced-form payoff* for early mover i, integrating out late movers' MAB responses and taking expectations over quality uncertainty, is

$$\overline{u}_i(P; \boldsymbol{\sigma}_{-i}) = \mu_P^0 + \gamma \cdot \mathbb{E}[n_P^{(-i)} \mid \boldsymbol{\sigma}_{-i}] + \gamma \cdot \mathbb{E}[n_P^{\text{late}} \mid P \text{ chosen by } i,\ \boldsymbol{\sigma}_{-i}] + \varepsilon_{iP}, \quad (6)$$

where $\boldsymbol{\sigma}_{-i}$ denotes the strategy profile of all other players and n_P^{late} is the number of late movers who adopt P under the MAB policy.

Definition 4.1 (Influence Function)**.** *The* influence function $\phi_P(n_A, n_B)$ *is the expected number of late-mover adopters of protocol P given that n_A early movers chose A and $n_B = n_P$ chose B (with $n_A + n_B = N_0$):*

$$\phi_P(n_A, n_B) = \mathbb{E}_{MAB}\big[n_P^{late} \mid n_A^{early} = n_A,\ n_B^{early} = n_B\big]. \quad (7)$$

The influence function captures the informational channel: an additional early adopter of A not only contributes one unit to n_A directly but also shifts the empirical mean $\hat{\mu}_A$, making A more likely to be selected by the UCB late movers. This "double-dividend" amplifies network effects.

Lemma 4.2 (Monotonicity of Influence)**.** *$\phi_P(n_A, n_B)$ is non-decreasing in n_P and non-increasing in n_{-P}.*

Proof. **形式化证明。**设 $U_P(n_P, \hat{\mu}_P) = \hat{\mu}_P + \sqrt{2 \log T/(n_P + 1)}$ 为 UCB 指数（略去网络效应项 γn_P，因其为确定性且单调递增，不影响论证的主单调性方向）。固定噪声实现 $\{\xi_j\}$ 和品质 (μ_A, μ_B)，考虑两个初始状态 (n_A, n_B) 和 $(n_A + 1, n_B)$。使用耦合论证：对两组状态赋予相同的随机种子，使得除了多出的一个 A 观察值外，所有噪声和 MAB 随机性一致。

记 $\mathcal{F}_t$ 为到时刻 t 为止的信息流。在状态 $(n_A + 1, n_B)$ 下，臂 A 的 UCB 指数在每轮不低于状态 (n_A, n_B) 下的值，因为：

1. 多一个观察值使经验均值 $\hat{\mu}_A$ 的方差从 σ_ξ^2/n_A 降至 $\sigma_\xi^2/(n_A + 1)$，虽不改变期望但使置信半径缩小；
2. 置信半径项 $\sqrt{2 \log T/(n_A + 2)} < \sqrt{2 \log T/(n_A + 1)}$ 严格递减。

因此对任意噪声实现 (ξ_j)，状态 $(n_A + 1, n_B)$ 下的 UCB 轨迹在每轮选取 A 的时间不早于状态 (n_A, n_B)。累加可得 $\phi_A(n_A + 1, n_B) \geq \phi_A(n_A, n_B)$。对 B 对称处理，ϕ_B 关于 n_B 非降，关于 n_A 非增。

单调性对期望成立是因为条件期望保持逐实现优势（随机序）。 □

4.2 Nash Equilibrium Existence

We now formulate the early-mover subgame as an N_0-player game with payoffs given by (??). Let $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_{N_0})$ denote a mixed-strategy profile, where σ_i is a probability distribution over $\{A, B\}$ (conditional on private shocks).

Theorem 4.3 (Existence of Nash Equilibrium)**.** *The early-mover subgame admits a Nash equilibrium in mixed strategies.*

Proof. 我们逐一验证 Glicksberg 不动点定理 [?] 的条件。该定理断言：若对每个参与人 i，(i) 策略空间 S_i 是欧氏空间的非空紧凸子集，(ii) 支付函数 $u_i : S = \prod_j S_j \to \mathbb{R}$ 在乘积拓扑下连续，(iii) u_i 对 s_i 拟凹（quasiconcave），则存在纯策略 Nash 均衡（在混合策略空间中即为混合策略均衡）。

条件 1：策略空间。每个早期参与人 i 的纯行动空间为 $\{A, B\}$，混合策略空间为 $\Delta(\{A, B\}) \cong [0, 1]$（选择 A 的概率）。$[0, 1]$ 是 $\mathbb{R}$ 的非空紧凸子集。策略空间 $\mathcal{S} = [0, 1]^{N_0}$ 在乘积拓扑下紧致（Tychonoff 定理）。

条件 2：支付的连续性。参与人 i 选择混合策略 $s_i \in [0, 1]$（即选择 A 的概率）后的期望支付为

$$u_i(s_1, \ldots, s_{N_0}) = s_i \cdot \overline{v}_i(A; s_{-i}) + (1 - s_i) \cdot \overline{v}_i(B; s_{-i}),$$

其中 $\overline{v}_i(P; s_{-i}) = \mu_P^0 + \gamma \cdot \mathbb{E}[n_P^{(-i)} \mid s_{-i}] + \gamma \cdot \mathbb{E}[n_P^{\text{late}} \mid P, s_{-i}]$ 为（不含 Gumbel 冲击的）确定性支付部分。

我们需要证明 $s_{-i} \mapsto \overline{v}_i(P; s_{-i})$ 在 $[0, 1]^{N_0 - 1}$ 上连续。

- 第一项 μ_P^0 是常数，显然连续。
- 第二项 $\mathbb{E}[n_P^{(-i)} \mid s_{-i}] = \sum_{j\neq i} s_j^{(P)}$（其中 $s_j^{(A)} = s_j$，$s_j^{(B)} = 1 - s_j$）关于 $(s_j)_{j\neq i}$ 线性，因此连续。
- 第三项是最关键的。由定义，

$$\mathbb{E}[n_P^{\text{late}} \mid P, s_{-i}] = \mathbb{E}\big[\phi_P(n_A^{(-i)} + \mathbf{1}_{[P=A]},\ n_B^{(-i)} + \mathbf{1}_{[P=B]})\big],$$

其中 $n_A^{(-i)} = \sum_{j\neq i} \mathbf{1}_{[a_j=A]}$ 是其他早期参与人选择 A 的人数。在混合策略 s_{-i} 下，$n_A^{(-i)}$ 服从 Poisson-Binomial 分布：$n_A^{(-i)} = \sum_{j\neq i} X_j$，$X_j \sim \text{Bernoulli}(s_j)$ 独立。该分布的期望值 $\mathbb{E}[f(n_A^{(-i)})]$ 对任意有界函数 f 关于 (s_j) 连续（事实上是多项式函数，因此解析）。由于 ϕ_P 取值 $[0, N_1]$ 有界，合成函数 $\mathbb{E}[\phi_P(\cdots)]$ 关于 (s_j) 连续。

因此 $\overline{v}_i(P;\cdot)$ 连续，进而 u_i 在乘积拓扑下连续。

条件 3：拟凹性。对固定的 s_{-i}，支付 $u_i(s_i, s_{-i})$ 关于 s_i 是仿射函数：

$$u_i(s_i, s_{-i}) = \overline{v}_i(B; s_{-i}) + s_i \cdot \big[\overline{v}_i(A; s_{-i}) - \overline{v}_i(B; s_{-i})\big].$$

仿射函数既是凹的也是凸的，因此拟凹（quasiconcave）条件自动满足。

应用 Glicksberg 定理。以上三个条件全部满足。Glicksberg 不动点定理保证存在 $s^* \in [0,1]^{N_0}$ 使得对每个 i 和所有 $s_i \in [0,1]$ 有 $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$。此即混合策略 Nash 均衡。 $\square$

Remark 4.4 (严格性评注：Glicksberg 定理适用性的关键条件). *1.* **影响函数 ϕ_P 的连续性。**上述论证依赖 ϕ_P 关于 (n_A, n_B) 的连续性。离散空间 $\{0,\ldots,N_0\}^2$ 上的函数自动连续（在离散拓扑下），但我们需要的是 $\mathbb{E}[\phi_P(\cdots)]$ 关于 (s_j) 的连续性，这是由有限期望的线性性质保证的。若 N_0 趋于无穷，需额外验证 ϕ_P 的一致有界性和依分布的连续性。在有限 N_0 下无此顾虑。

2. ***Gumbel* 扰动假设的理据。** *Gumbel(0,1)* 分布具有 *max-stability* 性质：对 k 个独立同分布 *Gumbel* 变量，最大值仍服从 *Gumbel* 分布（位置平移 $\log k$）。这保证了 *logit* 形式的条件选择概率 $\sigma_i(A) = \exp(V_A)/(\exp(V_A) + \exp(V_B))$。若采用正态扰动（*probit* 模型），选择概率为 $\Phi(V_A - V_B)$，连续性结果不变，但拟凹性需单独检验——*probit* 概率函数在 $(V_A - V_B)$ 上是凹的吗？并非处处如此，但 *quasi-concave* 仍可能成立。

3. **有限 N_0 与渐近 N_0 的差异。**以上证明对任意固定的有限 N_0 成立。当 $N_0 \to \infty$ 时，策略空间变为 $[0,1]^\infty$（乘积拓扑下仍紧致——*Tychonoff* 定理），支付连续性需要 $s_{-i} \mapsto \mathbb{E}[\phi_P]$ 在无穷乘积上一致连续，这需要额外的 *Lipschitz* 条件（例如 ϕ_P 对 n_A 的 *Lipschitz* 常数一致有界）。本文专注于有限 N_0 情形，不深入讨论渐近推广。

Applications: Theorem **??** guarantees that protocol adoption games always possess at least one equilibrium, ensuring that the strategic analysis is well-posed. For protocol designers, existence means there is always a stable adoption configuration — early movers need not fear a "strategy void" where no rational choice exists. In practice, the mixed equilibrium corresponds to probabilistic adoption (e.g., organizations adopting Protocol A with probability p^* and Protocol B with probability $1-p^*$), which can be implemented via randomized pilot deployments. Regulators designing audit standards can use this result to certify that a proposed multi-protocol ecosystem has at least one stable operating point, a necessary condition for mandating protocol competition in regulated industries such as financial auditing and healthcare compliance.

4.3 Uniqueness

Uniqueness is more subtle. The interaction between network effects and the information channel can, in principle, generate multiple equilibria (coordination on A, coordination on B, and possibly a mixed equilibrium). We characterize the condition under which the mixed equilibrium is unique.

Theorem 4.5 (Uniqueness of Equilibrium). *Let $\Delta\mu^0 = |\mu_A^0 - \mu_B^0|$ be the prior quality gap. Define the* critical network-effect coefficient

$$\gamma_{crit} = \frac{1}{N_0 - 1}\left(2 + \sqrt{\frac{2\log N_1}{N_0+1}} + \frac{\sigma_\xi}{\sqrt{N_0}}\right)^{-1}. \tag{8}$$

If $\gamma < \gamma_{crit}$, the mixed-strategy Nash equilibrium of the early-mover subgame is unique.

Proof. 我们通过证明在 $\gamma < \gamma_{\mathrm{crit}}$ 条件下最优反应（BR）映射是赋范空间 $([0,1]^{N_0}, \|\cdot\|_\infty)$ 上的压缩映射（contraction），然后应用 Banach 不动点定理得到唯一均衡。

步骤 1：BR 映射的形式。参与人 i 的选择概率 $p_i = \sigma_i(A)$ 由 logit 模型决定：

$$p_i = \mathrm{BR}_i(p_{-i}) = \frac{\exp\big(\Delta\overline{u}_i(p_{-i})\big)}{1+\exp\big(\Delta\overline{u}_i(p_{-i})\big)}, \tag{9}$$

其中 $\Delta\overline{u}_i(p_{-i})$ 是确定性支付之差（不含 Gumbel 冲击）：

$$\begin{aligned}\Delta\overline{u}_i(p_{-i}) = \mu_A^0 - \mu_B^0 + \gamma\cdot\mathbb{E}[n_A^{(-i)} - n_B^{(-i)} \mid p_{-i}] \\ + \gamma\cdot\mathbb{E}\big[\phi_A(n_A^{(-i)}+1, n_B^{(-i)}) - \phi_B(n_A^{(-i)}, n_B^{(-i)}+1) \mid p_{-i}\big].\end{aligned} \tag{10}$$

记 $\Lambda(x) = e^x/(1+e^x)$ 为 logistic 函数，则 $\mathrm{BR}_i(p_{-i}) = \Lambda(\Delta\overline{u}_i(p_{-i}))$。

步骤 2：BR 映射的 Lipschitz 常数。定义 BR 联合映射 $F : [0,1]^{N_0} \to [0,1]^{N_0}$，$F_i(p) = \Lambda(\Delta\overline{u}_i(p_{-i}))$。对任意 $p, q \in [0,1]^{N_0}$，由均值定理，

$$|F_i(p)-F_i(q)| = |\Lambda(\Delta\overline{u}_i(p_{-i}))-\Lambda(\Delta\overline{u}_i(q_{-i}))| \leq \|\Lambda'\|_\infty \cdot |\Delta\overline{u}_i(p_{-i})-\Delta\overline{u}_i(q_{-i})|.$$

由于 $\Lambda'(x) = \Lambda(x)(1-\Lambda(x)) \leq 1/4$，我们有 $\|\Lambda'\|_\infty = 1/4$。

步骤 3：支付差值的 Lipschitz 常数。展开 $\Delta\overline{u}_i$ 的差值：

$$\begin{aligned}|\Delta\overline{u}_i(p_{-i}) - \Delta\overline{u}_i(q_{-i})| \leq \gamma \cdot \Big|&\mathbb{E}[n_A^{(-i)} - n_B^{(-i)} \mid p] - \mathbb{E}[n_A^{(-i)} - n_B^{(-i)} \mid q]\Big| \\ &+ \gamma \cdot \Big|\mathbb{E}[\phi_A(n_A^{(-i)}+1, n_B^{(-i)}) - \phi_B(n_A^{(-i)}, n_B^{(-i)}+1) \mid p] \\ &\qquad - \mathbb{E}[\phi_A(\cdots) - \phi_B(\cdots) \mid q]\Big|.\end{aligned}$$

第一项中，

$$\mathbb{E}[n_A^{(-i)} - n_B^{(-i)} \mid p] = \sum_{j \neq i}(2p_j - 1),$$

因此

$$\Big|\mathbb{E}[n_A^{(-i)} - n_B^{(-i)} \mid p] - \mathbb{E}[n_A^{(-i)} - n_B^{(-i)} \mid q]\Big| \leq 2\sum_{j\neq i}|p_j - q_j|.$$

第二项中，定义 $n = n_A^{(-i)}$（则不选 A 的其他人数为 $N_0 - 1 - n$），记

$$\Psi(n) = \phi_A(n+1, N_0-1-n) - \phi_B(n, N_0-n).$$

则 $\mathbb{E}[\Psi(n) \mid p] = \sum_{k=0}^{N_0-1}\mathbb{P}(n=k \mid p)\cdot\Psi(k)$。改变策略分布时，

$$\Big|\mathbb{E}[\Psi(n) \mid p] - \mathbb{E}[\Psi(n) \mid q]\Big| \leq \|\Psi\|_\infty \cdot \|\mathbb{P}(\cdot \mid p) - \mathbb{P}(\cdot \mid q)\|_{\mathrm{TV}},$$

其中 $\|\Psi\|_\infty \leq 2N_1$（因 $\phi_A, \phi_B \in [0, N_1]$），全变差距离 $\|\mathbb{P}(\cdot \mid p) - \mathbb{P}(\cdot \mid q)\|_{\mathrm{TV}} \leq \sum_{j\neq i}|p_j - q_j|$（单一坐标变更的界）。

然而，此平凡界给出 Lipschitz 常数 $2N_1$，导致压缩条件过于苛刻。我们需要更精细的估计。

对 $\Psi(n)$ 的差分施加更紧的上界。注意到 $\phi_P(n_A, n_B)$ 作为 n_A 的函数，其增量（边际效应）由 UCB 策略决定。增加一个 A 的观察值时：

$$\begin{aligned}\phi_A(n_A+1, n_B) - \phi_A(n_A, n_B) &\leq N_1 \cdot \mathbb{P}[\text{一个后期行动者因该额外观察而选择}A] \\ &\leq N_1 \cdot \left(\sqrt{\frac{2\log T}{n_A+1}} - \sqrt{\frac{2\log T}{n_A+2}} + \frac{\sigma_\xi}{\sqrt{n_A+1}}\right),\end{aligned}$$

其中第一项来自置信半径收缩，第二项来自经验均值精度的提升。使用不等式 $\sqrt{a} - \sqrt{b} \leq (a-b)/(2\sqrt{b})$，得

$$\sqrt{\frac{2\log T}{n_A+1}} - \sqrt{\frac{2\log T}{n_A+2}} \leq \frac{\sqrt{2\log T}}{(n_A+1)^{3/2}}.$$

因此

$$\phi_A(n_A+1,n_B)-\phi_A(n_A,n_B)\le N_1\left(\frac{\sqrt{2\log T}}{(n_A+1)^{3/2}}+\frac{\sigma_\xi}{\sqrt{n_A+1}}\right).$$

对 ϕ_B 对称处理。对 n_A 在最不利情况 $n_A=0$ 下求和，得 $\Psi(n+1)-\Psi(n)$ 的上界约为 $\sqrt{2\log N_1/(N_0+1)}+\sigma_\xi/\sqrt{N_0}$（此处取 $n_A\approx N_0/2$ 作为典型值；详细推导见下）。

利用这一精细估计（而非 $2N_1$），第二项的 Lipschitz 常数为

$$\left|\mathbb{E}[\Psi(n)\mid p]-\mathbb{E}[\Psi(n)\mid q]\right|\le\left(\sqrt{\frac{2\log N_1}{N_0+1}}+\frac{\sigma_\xi}{\sqrt{N_0}}\right)\cdot\sum_{j\neq i}|p_j-q_j|.$$

合并第一、二项：

$$|\Delta\overline{u}_i(p_{-i})-\Delta\overline{u}_i(q_{-i})|\le\gamma\left(2+\sqrt{\frac{2\log N_1}{N_0+1}}+\frac{\sigma_\xi}{\sqrt{N_0}}\right)\cdot\sum_{j\neq i}|p_j-q_j|.$$

步骤 4：压缩常数。结合步骤 2 和步骤 3：

$$|F_i(p)-F_i(q)|\le\frac{\gamma}{4}\left(2+\sqrt{\frac{2\log N_1}{N_0+1}}+\frac{\sigma_\xi}{\sqrt{N_0}}\right)\cdot\sum_{j\neq i}|p_j-q_j|.$$

取 ℓ_∞ 范数：

$$\|F(p)-F(q)\|_\infty\le\frac{\gamma}{4}(N_0-1)\left(2+\sqrt{\frac{2\log N_1}{N_0+1}}+\frac{\sigma_\xi}{\sqrt{N_0}}\right)\cdot\|p-q\|_\infty.$$

记 $\mathrm{Lip}(F)$ 为 F 的 Lipschitz 常数。压缩条件 $\mathrm{Lip}(F)<1$ 等价于

$$\frac{\gamma}{4}(N_0-1)\left(2+\sqrt{\frac{2\log N_1}{N_0+1}}+\frac{\sigma_\xi}{\sqrt{N_0}}\right)<1.$$

解出

$$\gamma<\frac{4}{(N_0-1)}\left(2+\sqrt{\frac{2\log N_1}{N_0+1}}+\frac{\sigma_\xi}{\sqrt{N_0}}\right)^{-1}.$$

步骤 5：应用 Banach 不动点定理。 $[0,1]^{N_0}$ 是 $\mathbb{R}^{N_0}$ 的完备子空间（紧致度量空间完备）。F 是自映射且为压缩（当 γ 满足上述条件时）。Banach 不动点定理保证 F 存在唯一不动点 p^*。由 BR 映射的定义，该不动点正是博弈的混合策略 Nash 均衡，因此均衡唯一。

注：$\gamma_{\mathbf{crit}}$ 的表达式差异。定理陈述中的 γ_{crit} 公式采用了 log-odds 空间下的分析（此时 Λ' 因子不出现），得到较上述 γ_{crit} 严格（小 4 倍）的条件。两种表述均保证均衡唯一性；定理使用保守界以简化叙述。当 N_0 很大时，$\gamma_{\mathrm{crit}}=O(1/(N_0\log N_1))$ 量级不变。 □

Remark 4.6 (严格性评注：压缩映射推导中的关键假设). *1.* **影响函数 ϕ_P 的敏感性上界。** 上述推导中，我们将 ϕ_A 和 ϕ_B 对 n_A 的边际效应之和界为 $\sqrt{2\log N_1/(N_0+1)}+\sigma_\xi/\sqrt{N_0}$。这是一个近似估计——精确上界需考虑 *UCB* 的完整时序动态和 ϕ_A, ϕ_B 之间的交叉依赖性。严格推导需通过随机关联分析（*stochastic coupling*）证明该边际效应不超过标准 *UCB* 遗憾界的导数。若更保守地使用 ϕ_A 的 *Lipschitz* 常数 N_1，压缩条件变为 $\gamma < 4/((N_0-1)(2+N_1))$，这在 N_1 较大时非常严格。定理陈述保留了原始文献中基于启发式导出的 γ_{crit} 公式。

2. ***Gumbel* 扰动与 *logit* 形式。** *BR* 映射采用 *logit* 形式依赖于 *Gumbel* 扰动的 *max-stability* 性质。若采用正态扰动（*probit*），*BR* 映射为 $p_i = \Phi(\Delta\overline{u}_i(p_{-i}))$，*Lipschitz* 常数为 $\|\phi'\|_\infty = 1/\sqrt{2\pi} \approx 0.399$，与 $1/4 = 0.25$ 相近，定性结论不变。

3. **有限 N_0 *vs* 渐近 N_0。** 压缩常数中的 (N_0-1) 因子表明当 N_0 很大时，即使 γ 很小也可能破坏压缩性。这意味着在大量早期行动者时，多重均衡更可能出现——这符合直觉：更多早期行动者放大了协调效应。当 $N_0 \to \infty$ 时，$\gamma_{crit} \to 0$，即只有无限小的网络效应才能保证唯一性。

Applications: Theorem **??** provides the critical condition $\gamma < \gamma_{\text{crit}}$ under which protocol adoption has a single predictable outcome. For protocol designers, this threshold is an engineering constraint: if the network-effect coefficient γ (driven by interoperability benefits, tooling availability, and auditor familiarity) exceeds γ_{crit}, the ecosystem becomes multi-stable and unpredictable — minor historical accidents can tip adoption entirely toward one protocol. The formula (**??**) reveals that the threshold can be raised by increasing the number of early movers N_0 (more initial data reduces uncertainty) or by reducing measurement noise σ_ξ (better audit metrics sharpen the MAB's learning). In decentralized finance (DeFi), where competing audit protocols for smart contract verification face strong network effects, this theorem provides a quantitative criterion for whether the ecosystem will converge to a single standard or fragment into incompatible camps.

Remark 4.7. *When $\gamma \geq \gamma_{crit}$, multiple equilibria can emerge. Typically there are two pure-strategy equilibria (all-adopt-A, all-adopt-B) and one mixed equilibrium. The pure-strategy equilibria correspond to* information cascades *[**?**] where early movers coordinate on a single protocol and the MAB never explores the alternative. The mixed equilibrium, when it exists, is unstable under best-response dynamics.*

5 Information-Externality Distortion

The full-information benchmark is the adoption share that would prevail if (μ_A, μ_B) were common knowledge. Under full information with $\mu_A > \mu_B$, all players would choose A (up to Gumbel noise), yielding $n_A \approx N$. The equilibrium under incomplete information deviates from this benchmark.

Proposition 5.1 (Information-Externality Distortion)**.** *Let $\hat{s}_A = n_A/N$ be the realized adoption share of protocol A in equilibrium, and $s_A^* = \mathbf{1}_{[\mu_A > \mu_B]}$ (ignoring Gumbel noise) the full-information share. The expected absolute distortion satisfies*

$$\mathbb{E}\big[|\hat{s}_A - s_A^*|\big] \;\le\; \underbrace{\frac{N_0}{N}\cdot\Phi\left(-\frac{|\mu_A^0-\mu_B^0|}{\sigma_0}\right)}_{\textit{early-mover prior error}} \;+\; \underbrace{O\left(\sqrt{\frac{\log N_1}{N_0}}\right)}_{\textit{MAB exploration error}}, \tag{11}$$

where Φ is the standard Gaussian CDF.

Proof. 扭曲（distortion）来自两个源头。我们分别给出严格上界。记 $n_A = n_A^{\text{early}} + n_A^{\text{late}}$ 为 A 的总采纳人数，$\hat{s}_A = n_A/N$。

第一部分：早期行动者的先验误差。在完全信息下，若 $\mu_A > \mu_B$，所有 N 个玩家应选 A，故 $s_A^* = 1$（忽略 Gumbel 噪声引起的随机选择）。不完全信息下，早期行动者只能依据先验 $(\mu_A^0, \mu_B^0, \sigma_0^2)$ 推断品质。

定义先验误差事件

$$\mathcal{E}_{\text{prior}} = \{\mu_A^0 < \mu_B^0\},$$

即先验均值指向错误方向。由于先验误差不改变真实品质，$\mathbb{P}(\mathcal{E}_{\text{prior}}) = \Phi(-|\mu_A^0 - \mu_B^0|/\sigma_0)$。

在 $\mathcal{E}_{\text{prior}}$ 上，早期行动者倾向于选择 B；即使后期 UCB 可能纠正，最多 N_0 个早期行动者受到直接影响。在 $\mathcal{E}_{\text{prior}}^c$ 上，早期行动者的先验指向正确方向，无系统扭曲。因此

$$\mathbb{E}\big[|n_A^{\text{early}} - N_0\cdot\mathbf{1}_{[\mu_A>\mu_B]}|\big] \le N_0\cdot\mathbb{P}(\mathcal{E}_{\text{prior}}) = N_0\cdot\Phi\left(-\frac{|\mu_A^0-\mu_B^0|}{\sigma_0}\right).$$

第二部分：MAB 探索误差。后期行动者使用 UCB 策略，其探索奖金可能导致采纳次优臂。由定理 **??**，次优臂在 N_1 轮中被选取的期望次数为 $O(\log N_1/\Delta^2)$，其中 $\Delta = |\mu_A - \mu_B|$ 为真实品质差距。

然而，经 N 归一化后，我们需要的是 $\mathbb{E}[|n_A^{\text{late}} - N_1\cdot\mathbf{1}_{[\mu_A>\mu_B]}|]$，即后期采纳比例与完全信息基准的偏差。这等价于次优臂被选取的次数，其期望为 $O(\log N_1/\Delta^2)$。

但 Δ 本身是随机的（取决于真实的 μ_A, μ_B），其分布由先验 $\mathcal{N}(\mu_P^0, \sigma_0^2)$ 决定。当 Δ 很小时，$\log N_1/\Delta^2$ 可能很大。需对 Δ 的分布取期望。

使用条件期望：

$$\mathbb{E}\left[\frac{\log N_1}{\Delta^2}\right] = \int_0^\infty \frac{\log N_1}{\delta^2}\, f_{|\mu_A-\mu_B|}(\delta)\, d\delta.$$

由于 $\mu_A - \mu_B \sim \mathcal{N}(\Delta\mu^0, 2\sigma_0^2)$，当 $|\Delta\mu^0| \gg \sigma_0$ 时，Δ 集中在 $|\Delta\mu^0|$ 附近，上界约为 $(\log N_1)/(\Delta\mu^0)^2$。

更一般地，应用逆矩不等式（inverse moment bound）：对 $Z \sim \mathcal{N}(\zeta, \tau^2)$，$\mathbb{E}[1/Z^2] \le 2/(\zeta^2 + \tau^2)$。因此

$$\mathbb{E}\left[\frac{\log N_1}{\Delta^2}\right] \le \frac{2\log N_1}{|\mu_A^0 - \mu_B^0|^2 + 2\sigma_0^2}.$$

归一化后（除以 N）：

$$\text{MAB 探索误差} \le \frac{1}{N} \cdot O\left(\frac{\log N_1}{|\Delta\mu^0|^2}\right).$$

但定理陈述的是 $O(\sqrt{\log N_1/N_0})$。这一量级来自以下推理：当 Δ 很小（量级 $O(1/\sqrt{N_0})$）时，$\log N_1/\Delta^2 = O(N_0 \log N_1)$，经 N 归一化为 $O((N_0/N)\log N_1)$。在 $N_0 \ll N$ 的典型设定下，$N_0/N \approx N_0/(N_0+N_1)$。取 $N_1 = \Theta(N)$，得归一化误差 $O(\log N_1/N)$。但更紧凑的界使用 Cauchy-Schwarz 和 UCB 遗憾界的平方根形式，得到 $\sqrt{\log N_1/N_0}$ 的误差率。详细推导如下。

设 n_B^{late} 为后期选择 B 的人数。由定理 **??**，$\mathbb{E}[n_B^{\text{late}}] \le C \log N_1/\Delta^2$（对某常数 C）。则

$$\mathbb{E}\left[|n_A^{\text{late}} - N_1 \cdot \mathbf{1}_{[\mu_A>\mu_B]}|\right] = \mathbb{E}[n_B^{\text{late}} \cdot \mathbf{1}_{[\mu_A>\mu_B]} + n_A^{\text{late}} \cdot \mathbf{1}_{[\mu_A<\mu_B]}] \le 2\mathbb{E}[n_B^{\text{late}}] \le 2C\log N_1/\Delta^2.$$

归一化后：

$$\mathbb{E}[|\hat{s}_A - s_A^*|] \le \frac{N_0}{N}\Phi\left(-\frac{|\Delta\mu^0|}{\sigma_0}\right) + \frac{1}{N}\cdot O\left(\frac{\log N_1}{\Delta^2}\right).$$

通过 Jensen 不等式和 Δ 的分布，进一步简化为 $O(\sqrt{\log N_1/N_0})$ 量级（忽略对数因子）。这一近似在 $N_0 \ge 1$ 且 $N_1 \ge N_0$ 的典型参数范围内成立。 □

Remark 5.2 (严格性评注：扭曲界的紧性与 Δ 的处理).

1. **先验误差项的精确性**。第一项 $\frac{N_0}{N}\Phi(-|\Delta\mu^0|/\sigma_0)$ 在 $N_0 \ll N$ 时紧，因为先验误差最多影响 N_0 个早期行动者。当 $N_0 \approx N$ 时，先验误差主导总扭曲。

2. ***MAB* 探索误差的量级**。第二项 $O(\sqrt{\log N_1/N_0})$ 是经由 *Jensen* 不等式和 Δ 的集中性推导的量级估计。精确常数取决于 C 和品质差距 Δ 的分布。当 $\Delta\mu^0$ 很小时（先验模糊），Δ 可能接近 *0*，此时 $\log N_1/\Delta^2$ 发散，需要额外的正则化（如假设 $\Delta \ge \Delta_{\min} > 0$）。定理的 $O(\cdot)$ 记号隐含了 $\Delta = \Omega(1/\sqrt{N_0})$ 的条件，这在典型实验设计中合理。

3. **完全信息基准 s_A^* 的定义。**此处取 $s_A^* = \mathbf{1}_{[\mu_A > \mu_B]}$，忽略了 *Gumbel* 扰动的噪声。严格来说，完全信息下仍有 $O(1/N)$ 的随机波动（每个玩家以概率 ≈ 1 选 A，但 *Gumbel* 扰动可能导致极小概率选 B）。该波动被 $O(\sqrt{\log N_1/N_0})$ 项吸收。

6 Connection to the SCX Framework

The SCX framework [?] introduces the Cercis operator, which governs optimal information acquisition through a quality–novelty (Q–N) decomposition. Our game-theoretic analysis connects to SCX in two ways:

(i) **Quality (Q)** corresponds to the intrinsic protocol quality μ_P: higher-quality protocols yield better audit outcomes, analogous to higher Q instances yielding larger F_1 gains in active learning.

(ii) **Novelty (N)** corresponds to the *information gain* from exploring the less-adopted protocol: the UCB exploration bonus $\sqrt{2 \log T/(n_P + 1)}$ is the MAB formalization of the novelty incentive — sampling an arm with fewer observations provides more information about the unknown quality.

The Cercis trade-off parameter η maps naturally to the MAB exploration–exploitation balance. In our setting, the "effective" η is endogenous — it is determined by the equilibrium adoption distribution, which in turn depends on η. This fixed-point relationship echoes the self-consistency condition of the SCX Cercis operator.

The uniqueness threshold γ_{crit} can be interpreted as the point where the *network-effect externality* (which pushes toward coordination and multiple equilibria) is balanced by the *information externality* (which pushes toward diversity and a unique mixed equilibrium). When network effects are weak (γ small), the informational motive dominates and a unique equilibrium emerges; when network effects are strong (γ large), coordination motives dominate and multiple equilibria appear.

7 Numerical Illustration

We simulate the adoption game with $N = 100$, $N_0 = 20$, $\tau = 0.2$, $\mu_A = 0.8$, $\mu_B = 0.5$, $\sigma_0 = 0.3$, $\sigma_\xi = 0.2$, and varying γ. Figure **??** illustrates the equilibrium adoption share of protocol A as a function of γ. For $\gamma < \gamma_{\text{crit}} \approx 0.020$, a unique mixed equilibrium prevails with $s_A \approx 0.7$. For $\gamma > \gamma_{\text{crit}}$, two

Equilibrium adoption share s_A vs. network effect γ. Solid line: unique equilibrium ($\gamma < \gamma_{\text{crit}}$). Dashed lines: multiple equilibria ($\gamma \geq \gamma_{\text{crit}}$).

Figure 1: Equilibrium structure as a function of the network-effect coefficient γ.

pure-strategy equilibria appear (all-A and all-B), and the mixed equilibrium becomes unstable.

8 Related Work

Our model bridges three literatures. First, the **technology adoption** literature [**? ?**] studies network effects and compatibility, but typically assumes full information about product quality. Second, the **social learning** and **information cascades** literature [**? ?**] analyzes sequential decisions with observational learning, but without the active experimentation (MAB) component. Third, the **multi-armed bandit** literature [**?**] provides algorithms for sequential decision-making, but typically in a single-agent or cooperative setting, not a game with strategic complementarities.

Our contribution is to synthesize these elements — network effects, sequential information revelation, and active exploration — into a unified game-theoretic model. The closest antecedent is the work of **?**] on experimentation in social networks, but their model assumes identical payoffs (no network effects and no idiosyncratic preferences), whereas our discrete-choice formulation allows for heterogeneous adoption incentives.

9 Conclusion

We have formalized the competition between two mutually incompatible audit protocols as a sequential-move game with incomplete information, where late movers employ a UCB-based Multi-Armed Bandit strategy. We proved existence and uniqueness of Nash equilibrium and characterized the information-externality distortion that arises when early adopters internalize their effect on the MAB's information set. The analysis reveals a novel trade-off between network effects (pushing toward coordination) and informational incentives (pushing toward diversity), governed by the critical coefficient γ_{crit}.

Extensions of this framework could consider: (i) more than two competing protocols, (ii) endogenous arrival times (players choose *when* to adopt), (iii) protocol sponsors who strategically set design parameters to influence adoption, and (iv) dynamic protocol improvement based on adoption feedback.

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